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Abstract

Agriculture is facing significant challenges due to fluctuating market prices, inconsistent crop quality assessment, supply chain inefficiencies, and limited access to real-time decision support for farmers. To address these issues, this project, **CropSight: Smart Crop Delivery with AI-Based Price and Quality Intelligence**, presents an intelligent agricultural platform that integrates Artificial Intelligence (AI), Machine Learning (ML), and Internet of Things (IoT) technologies. The system enables farmers to assess crop quality, predict market prices, and streamline crop delivery processes through a data-driven and automated approach, thereby enhancing profitability and reducing post-harvest losses.

The proposed framework utilizes image-based crop quality analysis to evaluate factors such as size, color, texture, and defects using deep learning models. Simultaneously, machine learning algorithms analyze historical market trends, weather conditions, demand patterns, and regional price variations to generate accurate crop price predictions. IoT sensors and cloud-based services provide real-time monitoring of storage and transportation conditions, ensuring that crops maintain their quality throughout the supply chain. The platform also incorporates explainable AI techniques to provide transparent insights into quality grading and pricing decisions.

Furthermore, CropSight features a smart delivery recommendation module that connects farmers, distributors, and buyers through an integrated digital marketplace. The system recommends optimal delivery routes, suitable buyers, and the best time for selling crops to maximize returns. By combining quality intelligence, price forecasting, and smart logistics, the proposed solution improves supply chain efficiency, promotes fair pricing, reduces wastage, and empowers farmers with actionable insights for sustainable and profitable agriculture.

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